Churn Sage: Leveraging Machine Learning and Explainable AI for Enhanced Customer Retention

Er. Disha Sharma[1][0000-0002-9011-7899] , Satyam Kumar[2][0009-0002-0463-971X], Chiranjeev Kumar Singh[3][0009-0003-5161-2488], Shivam Kumar[4][0009-0008-2553-032X], Anshul Gautam[5][0009-0002-8503-3030]

[1] Assistant Professor, Department of CSE, Chandigarh University, Mohali, Punjab
[2] Department of CSE, Chandigarh University, Mohali, Punjab
[3] Department of CSE, Chandigarh University, Mohali, Punjab
[4] Department of CSE, Chandigarh University, Mohali, Punjab
[5] Department of CSE, Chandigarh University, Mohali, Punjab
disha.e13241@cumail.in

Abstract. In today's competitive market, predicting customer churn is important for long-term business management. Churn Sage is a machine learning-based solution designed to predict is essential for maintaining long-term business success. Churn Sage is a machine learning-based solution developed to forecast customer attrition using Python and advanced predictive analytics. By leveraging historical customer data, the model identifies patterns and behaviors that indicate potential churn, utilizing many techniques, including decision trees, logistic regression, and gradient boosting to achieve accurate predictions.

The framework applies feature engineering to optimize model performance, while hyper parameter tuning enhances prediction precision. Moreover, Churn Sage integrates explainable AI (XAI) to provide clear understanding of the elements behind client attrition, allowing companies to make informed decisions on retention strategies.

This research demonstrates how predictive analytics can help businesses proactively retain customers by identifying at-risk segments, thus reducing turnover rates. Churn Sage offers a scalable and efficient approach to churn prediction, applicable across industries such as e-commerce and telecommunications.

Ke ywords: Machine learning, Python, Explainable AI (XAI), Predictive analytics, Retention strategies, Churn forecasting.

1 Introduction

In today’s quickly advancing commerce scene, holding clients is more challenging than ever. With an expanding number of choices accessible to shoppers, client dependability can no longer be taken for allowed. Companies are presently centering on

client maintenance as a key procedure to keep up long-term commerce victory. Investigate has appeared that securing a unused client perhaps it is five times more expensive than running an existing business, highlighting the importance of churn management.

Customer churn, or steady loss, alludes to the misfortune of clients over a given period. For businesses, particularly in competitive segments such as broadcast communications, e- commerce, and keeping money, churn straightforwardly impacts income and development. As a result, anticipating and relieving churn has gotten to be basic for organizations pointing to improve client fulfillment and loyalty.

This paper presents Churn Sage, a machine learning-driven system planned to precisely anticipate client churn utilizing Python and progressed prescient analytics.

The show leverages authentic client data—such as engagement history, acquiring behavior, and exchange patterns—to identify early signals of potential steady loss. By utilizing machine learning calculations like calculated relapse, choice trees, and slope boosting, Churn Sage guarantees a tall level of forecast accuracy. The primary objective of this research includes:

- **Development of Predictive Model**: To create a python based model for forecasting customer churn.
- **Utilization of Algorithm**: To apply logistic regression, decision trees and gradient boosting.
- **Performance based optimization:** To enhance the performance of model through feature engineering and hyper parameter tuning.
- **Integration of Explainable AI**: To Use the XAI in identifying churn factors and transparency improvement.
- **Insight Generation**: Give data driven insights of knowledge based retention strategies.
- **Scalability**: Illustration of applicability across various industries.

Furthermore, the system utilizes feature engineering include building to improve the predictive control of the show and applies hyper parameter tuning to refine the algorithm's execution. The integration of reasonable AI (XAI) gives straightforwardness into the decision-making prepare, permitting businesses to get it the key components driving client churn.

This think about investigates how proactive churn expectation models like Churn Sage can be sent to recognize at-risk clients, eventually permitting businesses to actualize opportune and focused on maintenance techniques, making strides in general client fulfillment and diminishing turnover.

2 Literature Review

Shahabikargar, M., Beheshti et. al. (2024) [1]: The study introduces an information base (Churn KB) about customer attrition to improve churn prediction by extracting insights from customer interactions, such as emails and social media. Using Generative AI and feedback validation, the Churn KB improves machine learning models for more accurate churn predictions.

Phumchusri, N., & Amornvetchayakul, P. (2024) [2]: This study focuses on For a SaaS inventory management firm in Thailand, four machine learning methods were investigated for customer churn prediction: logistic regression, support vector machines, decision trees (DT), and random forests. With a 94.4% recall and an 88.2% F1-score, the optimized decision tree model performed the best. Relevance of features analysis provided insights into churn behavior, helping the company enhance decision-making and marketing strategies.

Aaron, J., Mogul, I., Hamid, T., Iwendi, C., & Aamir, A. (2024) [3]: This study explores customer churn in grocery retail amidst the rise of online shopping, emphasizing its negative effects on corporate growth and reputation. Utilizing machine and deep learning models, it analyzes customer behavior and preferences to uncover insights into churn. The research proposes a framework for proactive churn management, aiming to improve profitability and operational efficiency while enhancing overall business growth.

Amornvetchayakul, P et. al. (2020, April) [4]: This paper investigates customer churn in a rapidly growing a high churn inventory management software-as-a-service firm in Thailand rates. It measures four machine learning algorithms: decision trees, random forests, support vector machines, and logistic regression. The Random Forest model had a recovery of 91.6% and an F1 score of 92.6% with the lowest error rate. Feature importance analysis offers insights into churn-related metrics, aiding the company in identifying at-risk customers and enhancing decision-making and marketing efforts.

Telang, A. (2022) [5]: This report outlines a study conductedat DeJong and Laan, examining using artificial intelligence to forecast client attrition using their customer data. It emphasizes that churn is a dynamic aspect of the customer journey and serves as a crucial performance indicator of a business's health, influenced by factors like customer satisfaction and loyalty. The study operationalizes partial churn by analyzing various demographic variables and the recency, frequency, and monetary significance of customer purchasing behavior. A binary logistic regression model is employed to clients that are at danger, enabling the development specifically focused retention tactics.

Çallı, L., & Kasım, S. (2022) [6]: This study focuses on customer churn analysis in a B2B SaaS company, addressing the emerging topic of churn in this sector. Various

machine learning algorithms evaluated ten features, with the random forest algorithm yielding the best results. The analysis highlighted that the number of products and customer features significantly influence churn prediction.

Forghani Dehnavi, S. M., Rajabzadeh Ghatari, A., Azar, A., & Khadivar, A. (2022) [7]: This study explores customer churn and retention strategies using a systematic literature review, examining organizational factors, customer relationship management, and employee engagement. It introduces an algorithm using deep learning to forecast customer attrition, showcasing its effectiveness in accurately determining which clients are most likely to leave. The prototype also reveals significant factors influencing churn, providing valuable insights for organizations to enhance retention efforts.

Rujbr, V. (2016) [8]: This thesis discusses enhancing company profits through improved customer retention. It demonstrates how to utilize sales data for the automated early detection of potentially defecting customers, allowing targeted CRM interventions. The system also identifies rapidly growing customers, prioritizing them based on their behavior and strategic importance, enabling more effective CRM activities.

Grönros, L., & Janér, I. (2018) [9]: This thesis examines how Mr. Green, a leading online game provider in Europe, can predict customer churn using 24 hours of initial player data. Various Logistic regression, random forest, and other supervised machine learning models linear discriminant analysis, were evaluated. The best results came from a voting ensemble method, achieving an accuracy of 75.94% with specific weightings for each model.

Ahmad, A. K., Jafar, A., & Aljoumaa, K. (2019) [10]: This study addresses customer churn in the telecom industry by developing a predictive model using big data platforms and machine learning algorithms. To improve performance, the model integrates Social Network Analysis (SNA) features and creative feature engineering. from an AUC of 84%to 93.3%. Utilizing a large dataset from SyriaTel, the model was tested with four algorithms, with for optimal results, use Extreme Gradient Boosting (XGBoost). This model assists telecom operators in identifying customers most likely to churn, facilitating targeted retention strategies.

Khodabandehlou, S., & Zivari Rahman, M. (2017). [11]: This study presents a six-stage predictive framework for customer churn in business. It identifies key variables, such as number of items, returns, discounts, and distribution time, alongside RFM metrics, to improve churn prediction accuracy. The research finds that boosting models outperform the most accurate models are artificial neural networks (ANN), followed by basic and bagging models. The report emphasizes the need of putting client retention tactics into practice like prizes and quick deliveries.

Agrawal, S., Das, A., Gaikwad, A., & Dhage, S. (2018, July).[12]: This paper presents a client churn figure appear for a telecom dataset utilizing a significant learning ap-

proach. A multilayered neural organize was made to make a non-linear classification appear that predicts churn based on client, support, utilization, and pertinent highlights. The appear calculates the likelihood of churn though recognizing key affecting components. With an exactness of 80.03%, the appear enables companies to not as it were predicting client churn but as well analyze the components contributing to it. This information grants businesses to actualize strategies to address these factors and decrease churn effectively.

Ullah, I., Raza, B., Malik, A. K., Imran, M., Islam, S. U., & Kim, S. W. (2019). [13]: This paper proposes a churn forecast demonstrate for the telecom segment that combines classification and clustering strategies. Irregular Timberland accomplished 88.63% precision in distinguishing churn clients. The demonstrate employments highlight determination and categorizes churners into bunches for focused on maintenance offers. By recognizing key churn components, businesses can improve client maintenance and make strides showcasing techniques.

Çelik, O. et. al. (2019) [14]: This review examines customer churn analysis using various apparatus.

In addition to deep learning and the Cox proportional hazard model, learning techniques such artificial neural networks (ANN), decision trees, support vector machines (SVM), naive Bayes, k-NN, and XG Boost are used profound understanding offers larger success rates in complicated systems, even while time-related event prediction using machine learning techniques is still effective like customer churn.

Rahman, M. et. al. (2020, November). [15]: This paper proposes a procedure for forecasting bank customer attrition utilizing machine learning methods, such as Random Forest, SVM, KNN, and Decision Tree classifiers. After examining consumer behavior and feature selection, the Random Forest model with oversampling achieved the highest accuracy.

Dalvi, P. K., Khandge, S. K., Deomore, A., Bankar, A., & Kanade, V. A. (2016, March). [16]: Client churn expectation in the telecom industry has picked up noteworthy consideration due to showcase immersion and furious competition. This think about centers on anticipating client churn utilizing information mining and machine learning methods, such as calculated relapse and choice trees.

By analyzing telecom information, the ponder points to distinguish potential churners and offer assistance companies actualize successful maintenance procedures, comparing the proficiency of the calculations.

Umayaparvathi, V., & Iyakutti, K. (2017). [17]: The telecom market faces fierce competition, focusing on retaining customers rather than expanding the customer base. Customer churn prediction models depend heavily on selecting relevant customer attributes, which is often tedious and dataset- specific. This study explores deep learn-

ing for churn prediction, developing three neural network architectures. Results show deep learning models perform comparably to traditional models without manual feature selection.

Nagaraj, P. et. al. (2023, January). [18]: Customer retention is vital, especially in sectors like telecom and insurance where customer churn leads to increased costs and reduced margins. This paper highlights the importance of predicting churn through CRM systems, leveraging data mining and predictive analysis to identify patterns and reduce attrition rates.

Jajam, N., Challa, N. P., Prasanna, K. S., & Deepthi, C. V. S. (2023). [19]: This study proposes an advanced customer churn Stack bidirectional long short-term memory (SBLSTM) and RNN are used in this insurance industry-specific prediction (CCP) model, which is improved via the Arithmetic Optimization Algorithm (AOA). For the purpose of hyperparameter adjustment, the Improved Gravitational Search Algorithm (IGSA) is included model achieved superior accuracy of 97.89% and 97.67% across datasets.

Srinivasan, R. et. al. (2023, January). [20]: This paper explores various machine learning algorithms for predicting customer turbulence in the telecom industry. The Random Forest model in conjunction with SMOTE- ENN achieved the best results, with a 95% prediction accuracy based on the F1-score.

3 Methodology

3.1 Data Source:

The dataset for these inquiries about was sourced from an e- commerce stage, containing client behavior and exchange subtle elements for churn forecast. Information cleaning included evacuating unimportant highlights like Customer ID and taking care of lost values. Standard Scaler was utilized to normalize numerical factors, whereas categorical factors were changed over utilizing Name Encoding. These preprocessing steps guaranteed that the information was appropriately designed for preparing the machine learning models.

3.2 Model Selection:

Several machine learning models, such as Random Forest, Gradient Boosting, AdaBoost, XGBoost, Decision Trees, Gradient Regression, and LightGBM, were selected for churn prediction. A Voting Classifier was also used to Combine forecasts from many classifiers to improved accuracy.

3.3 Tuning hyperparameters and training models:

An 80-20 split of the data was made train-test ratio. Hyperparameters were optimized using GridSearchCV with 5- fold cross-validation in order to maintain strong model performance and avoid overfitting.

The selection of the above mentioned particular machine learning models in (3.2) and (3.3) for churn prediction which is often based on their strengths and suitability for the problem in this paper. These models are selected for handling complex pattern, high-dimensional data and providing robust prediction.

3.4 Measures of Evaluation:

To provide a fair evaluation of the models, accuracy, precision, recall, and F1-score were used customer churn predictions. As these metrics helped account for the two false positives and false negatives. Additionally, confusion matrices and ROC curves were plotted in order to offer a deeper analysis among the classification execution for each prototype.

Correctness: Precision is the proportion among the number of rectify forecasts (both genuine advantages and real drawbacks) to the total amount of expectations. It measures how frequently the show is rectify in foreseeing both churned and non-churned clients. Whereas exactness is a great common degree, it may not continuously give the most important result for imbalanced datasets, where one course (non-churned clients) rules.

$$\text{Accuracy} = \frac{TP+TN}{TP+FP+TN+FN}$$

Where:

- True Positives are called TP,
- True Negatives are called TN,
- False Positives are called FP,
- False Negatives are called TP.

In customer when predicting churn, precision can be misleading when the model favors the majority class (non- churn). Measures like recall, F1-score, and accuracy provide a more detailed assessment of a model's functionality.

Precision: Exactness measures that extent of genuine positives among all positive expectations. It is a pointer of how numerous clients anticipated to churn really churn. A higher accuracy shows a lower false-positive rate.

$$\text{Precision} = \frac{TP}{TP+FP}$$

Remind (Sensitivity): Recall, often called sensitivity, quantifies the percentage of correctly identified true positives. In churn prediction, it reflects how many actual churned customers the model accurately predicted.

$$\text{Recall} = \frac{TP}{TP+FN}$$

Elevated recollection is crucial when predicting churn to minimize false negatives, ensuring most at-risk customers are accurately identified.

The F1-Score: The F1-score represents the consonant unfeeling exactness and survey, giving a balanced a solitary measure. It's especially important when altering unfaithful positives and unfaithful negatives is imperative for illustrate evaluation.

$$\text{F1-Score} = 2 \text{ x } \frac{Precision \times Recall}{Precision + Recall}$$

The unbalanced datasets, the F1-score is useful, giving more focus to the minority class.

The curve representing the receiver operating characteristic: The formula used to determine the area under the ROC curve (AUC) is:

$$\text{AUC} = \int_0^1 TPR(FPR)d(FPR)$$

Where:

- TPR = True Positive Rate (Remind),
- FPR = False Positives Rate.

Confusion Matrix: The disarray network outwardly speaks to show execution, appearing true/false positives and negatives. An adjusted network recommends the show decently predicts both classes without inclination.

The structure of the confusion matrix is as follows:

Type of Churn	Predicted Churn(Positive)	Predicted Non-Churn(Negative)
Actual Churn	*TP*	*FN*
Actual non-churn	*FP*	*TN*

The matrix of misunderstanding enables calculation of metrics like precision and recall, highlighting model errors and guiding performance improvements.

AUC and the ROC Curve: The ROC curve, which plots the true positive rate versus the false positive rate at different thresholds, shows how well a model can differenti-

ate between consumers who have churned and those who have not. A performance curve that is closer to the upper left corner is preferable.

Model performance is measured by the AUC (Area under the ROC Curve), which goes from 0.5 for random guessing to 1.0 for perfect accuracy. An elevated AUC signifies superior model effectiveness in distinguishing between classes.

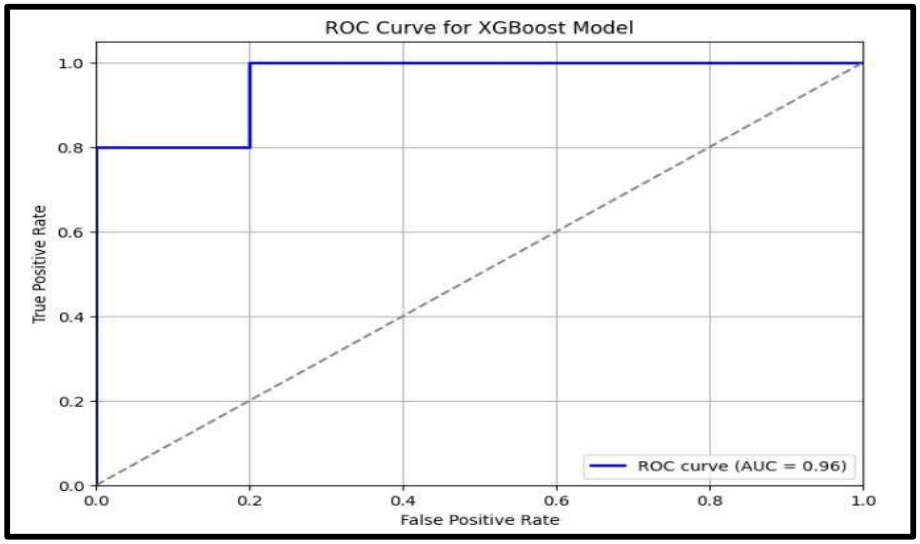


Fig. 1. ROC Curve for XGBoost Model on Churn Prediction Dataset

3.5 Implementation:

Experiments were conducted using Python 3.7, leveraging libraries like Seaborn, Matplotlib, and Pandas for data processing; Scikit-learn for visualization for model training and evaluation. Gradient boosting models were implemented using XGBoost, LightGBM, and CatBoost, known for their strong predictive capabilities.

Pandas: Offering Data Frames and Series for structured data, Pandas is a potent Python toolkit for data manipulation and analysis, along with features for cleaning, aggregation, and time series.

Seaborn: Seaborn is a Visualization toolkit built on top of Matplotlib that makes it easier to create visually appealing and educational statistics visuals.

It provides functions to create various plots, including:

- Heatmaps: Useful for visualizing correlations and data patterns.

- Boxplots: Ideal for visualizing distributions and detecting outliers.
- PairPlot: Great for examining relationships between multiple variables.

Matplotlib: Matplotlib is a core Python plotting library that allows for static, animated, and interactive visualizations, offering extensive customization and often used with Pandas and Seaborn.

Scikit-learn: It offers coordinate and effective insubordinate for information mining and information examination, including:

- **Classification:** Calculations for foreseeing categorical comes around (e.g., churn prediction).
- **Regression:** Strategies for expecting nonstop outcomes.
- **Clustering:** Methods for gathering comparable information points.
- **Model Assessment:** Insubordinate for looking over show up execution through estimations like exactness, exactness, review, and F1-score.

XGBoost: LightGBM is an angle boosting system that utilizes a histogram-based calculation, improving preparing speed and lessening memory utilization.

- **Efficiency:** Handles large datasets effectively with faster training times compared to traditional methods.
- **Accuracy:** Competes with other models like XGBoost in terms of predictive performance.
- **Flexibility:** Supports both categorical and numerical features natively.

CatBoot: A gradient booster is called CatBoost library which excels with categorical features, automatically handling them for easier implementation in relevant scenarios.

- **Robustness**: Handles overfitting through advanced regularization techniques.
- **Automatic Categorical Handling**: Processes categorical features without extensive preprocessing.
- **High Performance**: Often delivers competitive results with minimal tuning.

3.6 Model Validation and Testing:

After training and hyperparameter tuning, each model underwent rigorous validation with the testing dataset to evaluate generalization to unseen data. To ensure consistency, this measurement was repeated several times and the averages of metrics such as precision, recall, accuracy, and F1 were taken. Also, the area under the curve (AUC) and receiver operating characteristic (ROC) curves were measured.

Classification performance, with high AUC values (nearer 1) signifying successful differentiation between non-churned and churned clients.

3.7 Ensemble Learning Approach:

To enhance predictive performance, ensemble methods were utilized, combining multiple models for a more accurate final prediction. The Voting Classifier aggregated outputs from well-performing models like SVM, Gradient Boosting, Logistic Regression, and Random Forest, employing majority voting to reduce variance and bias. Additionally, soft voting was implemented, averaging predicted probabilities from each model to make the final decision. This approach prioritizes classifiers with stronger predictive performance, further improving overall accuracy and robustness in predictions.

3.8 Feature Importance Analysis:

A study of feature significance was carried out in order to understand the model and determine the main elements impacting customer attrition. Making use of models like Random Forest and Gradient Boosting, which have built-in feature importance mechanisms, we discovered that factors like the quantity of transactions, time spent on the platform, and client age significantly impact churn likelihood. This analysis provides valuable insights on what causes turnover, helping to inform methods for customer retention and improved service offerings.

3.9 Comparing with Baseline Model:

The machine learning models were compared to baseline models like Logistic Regression to assess improvements in predictive power. While Logistic Regression offers acceptable performance, gradient boosting and random forest models, among others demonstrated significant enhancements in precision and recall, indicating a greater capability to accurately predict consumers who may leave.

3.10 Computational Efficiency:

To address the computational intensity of training large ensemble models like XGBoost and LightGBM, techniques such as parallel processing and early stopping were used to minimize training times while maintaining accuracy, ensuring the solution remains scalable for larger datasets.

3.11 Machine Learning Classifier:

Decision tree classifier is a type of map-like model in which the anticipated class labels are shown by leaf nodes, while inside nodes represent judgments made on the basis of certain characteristics. The model attempts to successfully divide the classes by recursively dividing the data into subgroups using feature-based judgments. The quality of these splits is assessed using metrics like Gini impurity or entropy, which measure the reduction in uncertainty and maximize information gain. The main char-

acteristics of decision trees are open-mindedness and efficient processing of numerical and categorical data.

3.12 SVMs, or Support Vector Machines:

Support vector machines (SVM) are a potent classification technique that maximizes the margin between classes to choose the best hyperplane for dividing them in a feature space. The support vectors for each class are the locations that are closest to the hyperplane. SVM can solve classification and mismatch problems by transforming data into a high-dimensional space using kernel functions.

This makes it especially helpful when working with high- dimensional data and datasets that have clear margin limits. Because SVM concentrates on optimizing the margin, it is also good at handling outliers. The process of choosing the best algorithm usually include testing out many solutions and assessing how well they work given the particular dataset and challenge.

3.13 Regression using the logistic method:

Based on the properties of the input, one may use a technique called logistic regression to forecast the probability that an instance would fall into a certain class, like pneumonia or normal. It is ideal for problems involving binary classification because it uses the logistic function to convert output values into probabilities between 0 and 1.

3.14 Random Classifier:

A rudimentary machine learning model known as a Random Classifier, or Random Baseline, allocates class labels to instances entirely at random; it does not make use of learning or pattern recognition. It frequently serves as a standard by which more sophisticated models are measured for efficacy.

4 Results

4.1 Model Performance Evaluation:

Here, we have showcased the performance outcomes for all machine learning models running during the learning process, accuracy, precision, recall, F1 score, and ROC-AUC are the metrics used to evaluate a model to better understand its performance.

Table 1. Show results for models tested in the experiment, including Support Vector Machine (SVM), CatBoost, XGBoost, LightGBM, Decision Tree, Random Forest, Gradient Boosting, K nearest neighbor (KNN), and join the voting group.

4.2 Analysis of Results:

The study's victory in anticipating client churn highlights the potential for broader applications of machine learning in trade decision-making. Future work will point to make strides the model's adaptability, completing it reasonable for instantaneous examination and extensive datasets. Consolidating profound learning strategies, cross breed models, and robotized include building can assist optimize execution and versatility.

Logistic Regression and Decision Tree, though producing reasonable results, performed lower than ensemble and boosting models due to their simpler structure and limited ability to capture data complexity.

Table 1. Accuracy of Different Models on Churn Prediction Dataset.

Model	Accuracy (%)
Stacking Classifier	97.33
Bagging Classifier Accuracy	94.84
Voting Classifier Accuracy	96.81
Tuned Stacking Classifier Accuracy	97.51

4.3 Discussion of Model Performance:

Ensemble models and boosting techniques excel in churn prediction, combining strengths of individual models like Random Forest and SVM for better performance.

4.4 Feature Importance and Interpretability:

Key churn predictors include transaction frequency, platform usage time, and recency of purchase, offering actionable insights for customer retention strategies.

4.5 Potential for Importance:

Improvements could involve data augmentation, advanced hyperparameter tuning, and exploring deep learning models for better accuracy and complexity handling.

4.6 Limitations:

Limitations include dataset representation and class disparity, which might affect the model's performance generalizability and robustness in predicting churn.

5 Conclusion and Future Directions

The application of machine learning techniques, including ensemble models and boosting algorithm has proven effective in predicting customer churn prediction. The Voting Classifier developed as the best-performing demonstrate, giving a strong and precise arrangement for this basic trade issue. This include significance investigation advertised profitable experiences into client behavior, which can advise future maintenance procedures. Future work will center on upgrading the model's execution through progressed strategies such as profound learning and advance information enlargement.

The study's victory in anticipating client churn highlights the potential for broader applications of machine learning in trade decision-making. Future work will point to make strides the model's adaptability, achieving it reasonable in the present examination as well as extensive datasets. Consolidating profound learning strategies, cross breed models, and robotized include building can assist optimize execution and versatility.

Expanding on future work, integrating real-time data pipelines and exploring unsupervised learning approaches could enhance the model's adaptability. Incorporating customer segmentation and behavior clustering might reveal hidden patterns to refine retention strategies. Additionally, employing explainable AI techniques would improve transparency, aiding businesses in interpreting the model's predictions and making data-driven decisions more effectively.

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